

Elementer Satır İşlemlerini Kullanarak Bir Matrisin Tersini Bulma

A tersinir bir matris olmak üzere A^{-1} şu şekilde bulunabilir:

$$\left(\underline{A} \mid \underline{I} \right) \xrightarrow[\text{satır işlemleri}]{\text{elementer}} \left(\underline{I} \mid \underline{A^{-1}} \right)$$

Öl:

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \text{ matrisinin tersini bulunuz.}$$

Cözüm:

1.yol:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \Rightarrow A^{-1} = \frac{1}{3} \begin{pmatrix} 3 & -2 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -2/3 \\ 0 & 1/3 \end{pmatrix} \#$$

2.yol

$$\left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 3 & 0 & 1 \end{array} \right) \xrightarrow{\frac{1}{3}R_2} \left(\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & 1 & 0 & 1/3 \end{array} \right)$$

$$\xrightarrow{-2R_2 + R_1} \left(\begin{array}{cc|cc} 1 & 0 & 1 & -2/3 \\ 0 & 1 & 0 & 1/3 \end{array} \right) \Rightarrow A^{-1} = \begin{pmatrix} 1 & -2/3 \\ 0 & 1/3 \end{pmatrix}$$

Öl:

$$A = \begin{pmatrix} 2 & 2 & 0 \\ 0 & -1 & 0 \\ 3 & 2 & 4 \end{pmatrix} \text{ matrisinin tersini bulunuz.}$$

$$\left(\begin{array}{ccc|ccc} 2 & 2 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 3 & 2 & 4 & 0 & 0 & 1 \end{array} \right) \xrightarrow{\frac{1}{2}R_1} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1/2 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 0 & 3 & 4 & 0 & 0 & 1 \end{array} \right)$$

$$\xrightarrow{\begin{array}{l} -3R_1+R_3 \\ -R_2 \end{array}} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & -1 & 4 & -3/2 & 0 & 1 \end{array} \right) \xrightarrow{R_2+R_3} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 4 & -3/2 & -1 & 1 \end{array} \right)$$

$$\xrightarrow{\frac{1}{4}R_3} \left(\begin{array}{ccc|ccc} 1 & 1 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & -3/8 & -1/4 & 1/4 \end{array} \right).$$

$$\xrightarrow{-R_2+R_1} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/2 & 1 & 0 \\ 0 & 1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & -3/8 & -1/4 & 1/4 \end{array} \right) \Rightarrow A^{-1} = \begin{pmatrix} 1/2 & 1 & 0 \\ 0 & -1 & 0 \\ -3/8 & -1/4 & 1/4 \end{pmatrix}$$

Not: Bu yöntem boyutları 3×3 'ten büyük matrislerin tersini bulmak için en uygun yöntemlerden biridir.

Not: Bir matris elemanter satır işlemleri ile ya birim matrise ya da sıfır satır içeren bir matrise kadar indirgenemez.

Not: $(A | I) \rightarrow (I | A^{-1})$ formuna gelmiyorsa A matrisi terslenemez matristir.
 singular
 kesi I olmuyorsa!

Ör:

$A = \begin{pmatrix} 3 & 6 \\ 1 & 2 \end{pmatrix}$ matrisinin tersini (varsa!) bulunuz.

Çözüm:

$$\left(\begin{array}{cc|cc} 3 & 6 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array} \right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 3 & 6 & 1 & 0 \end{array} \right)$$

$$\xrightarrow{-3R_1 + R_2} \left(\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -3 \end{array} \right) \Rightarrow \underline{A \text{ terslenemez!}}$$

$\neq I$